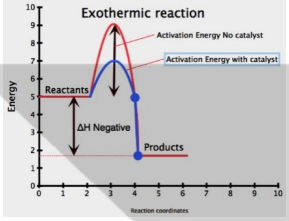
1.7. Catalyst

A catalyst is a substance that increases the reaction rate **by lowering the activation energy and is left unchanged at the end of the reaction.**

- More particles will have an energy greater than or equal to the activation energy, therefore successful collision rate increases, resulting in an increased rate of reaction
- For gaseous reactants, if the catalyst is solid metal, the catalyst provides a surface for the reaction.
- The larger the surface area of the metal catalyst, the larger the area for the reaction to take place; therefore higher the rate of reaction



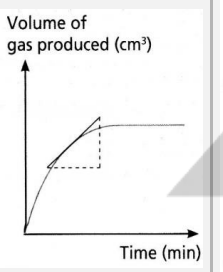
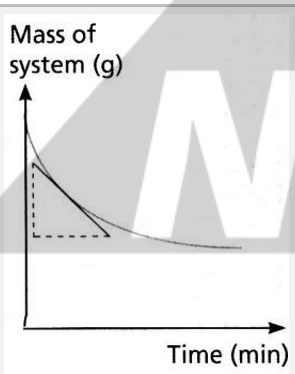
Enzymes Affecting Rate of Reaction

- Enzymes are biological catalysts that speed up reactions but remain chemically unchanged.
- Enzymes function best at optimum temperature and pH level; otherwise, they may denature and completely stop functioning.

1.8. Rates of Reaction Graphs

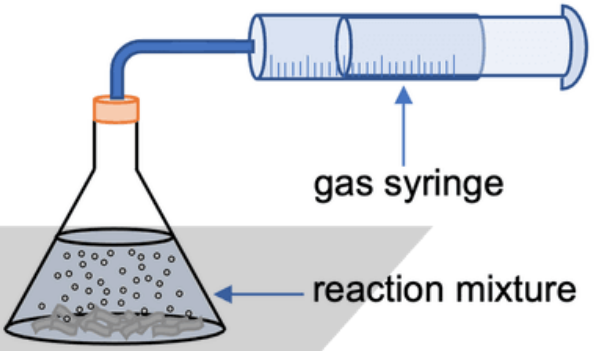
Interpreting graphs:

A graph with a steeper gradient at the beginning and reaching a horizontal gradient faster depicts a high rate of reaction.

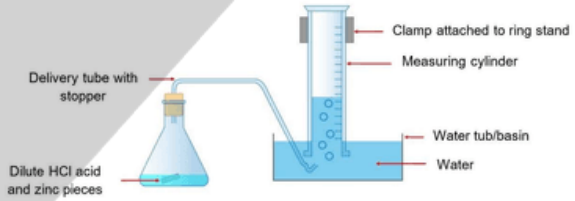
Gas Evolved	Mass Loss	Colour Change
		
If a gas evolves, measure the volume of gas produced per unit of time using a gas syringe	If a gas evolves, measure the loss in mass per unit time by placing it on a balance and then putting cotton wool on top to allow gas to pass but not enter	If a colour change, measure the time taken to turn cloudy

1.9. Evaluating Practical Methods

1. **Change in Mass method**
   - Not suitable for experiments where hydrogen gas is given off (because density decreases, too small mass)
2. Easier to set up the **gas syringe** and obtain reliable measurements



3. **An inverted Measuring cylinder** to collect gas over water is harder to set up.



1.10. Reversible Reactions and Equilibrium

The symbols of reversible reactions are: ⇌

- The products can then react with each other or decompose to form the reactant molecules.
- Two equations you should know:
- $CuSO_4 \cdot 5H_2O$ (blue) $\rightleftharpoons CuSO_4$ (white) $+ 5H_2O$
- $CoCl_2 \cdot 6H_2O$ (pink) $\rightleftharpoons CoCl_2$ (blue) $+ 6H_2O$

(anhydrous by heating; hydrated form by adding water)

Reversible Reaction in a closed system when equilibrium:

- Rate of forward reaction = rate of reverse reaction
- Concentrations of all reactants and products remain constant and are no longer changing

Equilibrium

- **Le Châtelier's Principle:** if conditions of equilibrium are changed, the position of the equilibrium moves to oppose change
- **Temperature:** Increase in temperature shifts the equilibrium in the endothermic direction. When the temperature is lowered, equilibrium shifts in the exothermic direction.
- **Pressure:** Pressure raised; equilibrium moves to the side with the fewest gas molecules. When pressure is lowered, equilibrium shifts to the side with most gas molecules.

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- **Concentration:** Decreasing reactant concentration or increasing product concentration; equilibrium moves to the reactant side. Increasing reactant concentration or decreasing product concentration; equilibrium shifts to the product side.
- **Catalyst:** The presence of a catalyst does **not** affect the equilibrium position but increases the rate at which equilibrium is reached. Therefore, it only increases the rate of the forward and backward reactions by the same amount.

Oxidising Agents are **Electron Acceptors** and Reducing Agents are **Electron Donor.**

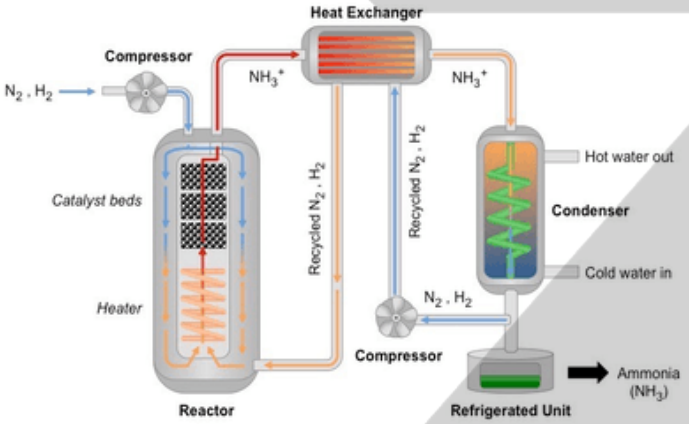
1.11. Haber Process

$N_2 + 3H_2 \rightleftharpoons 2NH_3$ (g) [Exothermic]

Materials (Reactants) in the Haber Process

Nitrogen: From the **air.**

Hydrogen: From the reaction between **Methane and Steam**



Conditions

Temperature: **450°C**

Pressure: **200 atm (20,000 kPa)**

Catalyst: **Iron catalyst**

Why are these conditions chosen in the Haber Process?

1. **Pressure is not extremely high** - (Although increased pressure is more yield, more pressure will require more expensive equipment, and it can explode).
2. **Optimum temperature** - (If lower temperature, more ammonia yield, however too low, the rate is uneconomical.

1.12. Contact Process

$2SO_2 + O_2 \rightleftharpoons 2SO_3$ (g) [Exothermic]

Materials (Reactants) in Contact Process

Sulfur Dioxide: **Burning sulfur or Roasting Sulfide ores**

Oxygen: From the air

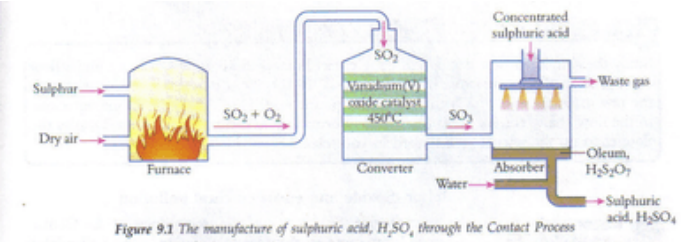


Figure 9.1 The manufacture of sulphuric acid, H_2SO_4 through the Contact Process

Conditions

Temperature: **450°C**

Pressure: **2 atm (200 kPa)**

Catalyst: **Vanadium (V) oxide catalyst**

Why are these conditions chosen in the Contact Process?

1. **Pressure is high** - (Increase pressure, more yield)
2. **Optimum temperature** - (If lower temperature, more sulfuric acid yield, however too low, the rate is uneconomical.

1.13. Redox

Redox: A simultaneous oxidation and reduction reaction

Oxidation	Reduction
Loss of electrons	Gain of electrons
Gain of oxygen	Loss of oxygen
Loss of hydrogen	Gain of hydrogen

NOTE: **Roman Numerals** next to an element's name are the **oxidation number of an element in a compound**, e.g. Iron (II) and Iron (III). The reaction product formula depends on this.

Reducing agents are oxidised, and oxidising agents are reduced.

Identifying Redox Reactions

1. **Oxidation State:**

The oxidation state highlights electron movement in a reaction

eg. CuO + Mg → MgO + Cu

Cu^{2+} + Mg → Mg^{2+}+ Cu [oxide is a spectator and is removed as it doesn't change its oxidation state]

Here Copper (II) is reduced while magnesium is oxidized; the reaction is redox

2. **Indicators:**
   - **Acidified Aqueous Potassium manganate (VII)** is a deep purple oxidising agent; when added to a reducing agent changes from purple to colourless.
   - **Aqueous Potassium iodide** is a reducing agent; when added to an oxidising agent, it changes from colourless to yellow-brown.

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